gGA Dimer Scans: U/t Dependence at $\Delta V = 0$ and $\Delta V = 2$

(Ghost Multiplicities $N_g = 0$ and $N_g = 2$, Direct-Minimization Solver)

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1 The $N_g = 0$ case (ordinary Gutzwiller)

At $\Delta V = 0$ the Gutzwiller equations of the dimer have a closed solution (`gGut_notes.pdf`, Sec. III D). With $V = -tR$ and $\lambda = \lambda^c = 0$ the embedding ground state gives, for $U \le U_c = 8t$ (Brinkman–Rice),

$$Z = R^2 = 1 - \Big(\frac{U}{8t}\Big)^2, \qquad E = -2t - \frac{U}{2} - \frac{U^2}{32t}, \qquad \langle n_\uparrow n_\downarrow\rangle = \frac{1}{4} - \frac{U}{32t}, \qquad P_{\rm ion} = \frac{1}{2} - \frac{U}{16t}, \quad (1)$$

and for $U \ge 8t$ only $R = 0$ exists: $Z = 0$, $E = -U$, $P_{\rm ion} = 0$. The quasiparticle Hamiltonian is $-tZ\,\sigma_x$, with levels $\pm tZ$ and orbitals $(1, \pm 1)/\sqrt{2}$ on the two states; each level carries the weight $2Z$, so the spectral function consists of two peaks, at $\pm tZ$ with weight $2Z$, that move to $\omega = 0$ and vanish at U_c (the gap $2tZ$ closes continuously). Here U_c is the value where the spectral function disappears. The closed forms agree with the numerical $N_g = 0$ solution of the code to 1.4×10^{-7} in Z and 10^{-8} in E and $\langle n_\uparrow n_\downarrow\rangle$ (script `ng0_analysis.py`).

At $\Delta V \neq 0$ particle–hole symmetry is broken, so $\lambda \neq 0$ and the embedding problem is 3×3 in the states $S = (a^\dagger_\uparrow d^\dagger_\downarrow - a^\dagger_\downarrow d^\dagger_\uparrow)/\sqrt{2}$, $D = a^\dagger_\uparrow a^\dagger_\downarrow$, $E = d^\dagger_\uparrow d^\dagger_\downarrow$ of the physical (a) and bath (d) orbitals. The site exchange combined with the particle–hole transformation relates the two fragments, $\phi_1 = (s, \epsilon, \delta)$

for $\phi_0 = (s, \delta, \epsilon)$ in the basis (S, D, E), so $Z_0 = Z_1$ and the problem has two variables, $c = \delta^2 + \epsilon^2$ and $m = \delta^2 - \epsilon^2$. With $r = \sqrt{c^2 - m^2}$:

$$E = -U(1-c) - 2\Delta V\, m - 4t\,\frac{(1-c)(c+r)}{\sqrt{1-m^2}}, \qquad Z = \frac{2(1-c)(c+r)}{1-m^2}, \qquad P_{\rm ion} = c, \qquad n = 1+m, \tag{2}$$

where n is the number of electrons on the favored site; the quasiparticle levels are $\pm tZ/\sqrt{1-m^2}$ with weight $2Z$ each ($\lambda = tZm/\sqrt{1-m^2}$), and at $\Delta V = 0$ ($m = 0$) all of this reduces to the formulas above. The stationarity conditions, $U\sqrt{1-m^2}\,r = 4t(c+r)(1-c-r)$ and $\Delta V(1-m^2)^{3/2} r = 2t(1-c)\,m\,(1-c^2-rc)$, are solved for (c, m); they reproduce the numerical $N_g = 0$ solution of the code (`scan_ng0/`) to 8×10^{-8} in E, 4×10^{-7} in Z and 5×10^{-8} in $P_{\rm ion}$ (script `ng0_dV_analytic.py`). Z vanishes for $c, m \to 0$ with $q = r/c$ fixed, where the two conditions give $U_c = 4t(1+q)/q$ and $\Delta V = 2t\sqrt{1-q^2}/q$, i.e.

$$U_c = 4t + 2\sqrt{4t^2 + \Delta V^2}, \tag{3}$$

which is $8t$ at $\Delta V = 0$, $9.66\,t$ at $\Delta V = 2$ and grows as $2\Delta V$ for large ΔV.

Quasiparticle orbitals. In the basis of the two states $H_{qp} = -\left(\begin{smallmatrix} \lambda & tZ \\ tZ & -\lambda \end{smallmatrix}\right)$ with $\lambda = tZ\,m/\sqrt{1-m^2}$, so the levels are $\mp E_q$, $E_q = tZ/\sqrt{1-m^2}$, and the orbitals (up to a global sign, chosen as that of the $N_g = 2$ HOMO and LUMO on the two states) are

$$|\mathrm{occ}\rangle = \left(\sqrt{\tfrac{1+m}{2}},\ \sqrt{\tfrac{1-m}{2}}\right), \qquad |\mathrm{virt}\rangle = \left(\sqrt{\tfrac{1-m}{2}},\ -\sqrt{\tfrac{1+m}{2}}\right), \tag{4}$$

i.e. the occupied orbital has the density $(1+m)/2$ on the favored state (0.945, 0.759 and 0.549 at $U/t = 0.1$, 4 and 8 for $\Delta V = 2$). At $\Delta V = 0$, $m = 0$ and the orbitals are $(1, \pm1)/\sqrt{2}$. Two limits are explicit: for $U \to 0$, $m \to \Delta V/\sqrt{t^2 + \Delta V^2}$, the non-interacting value (0.894 for $\Delta V = 2$; the solution gives 0.894 at $U/t = 0.01$); at U_c, $m \to 0$ and $Z \to 0$, so the orbitals become $(1, \pm1)/\sqrt{2}$ with levels at zero. For general $(U, \Delta V)$, m is not an explicit function of $(U, \Delta V)$; instead U and ΔV are explicit functions of (c, m),

$$U = \frac{4t\,(c+r)(1-c-r)}{\sqrt{1-m^2}\,r}, \qquad \Delta V = \frac{2t\,(1-c)\,m\,(1-c^2-rc)}{(1-m^2)^{3/2}\,r}, \qquad r = \sqrt{c^2-m^2}, \tag{5}$$

which is the stationarity condition solved for $(U, \Delta V)$; for given $(U, \Delta V)$ the pair (c, m) is its root (script `ng0_analytic.py`; the round trip $(U, \Delta V) \to (c, m) \to (U, \Delta V)$ closes to 10^{-6}).
In summary, $m = \delta^2 - \epsilon^2$ (half the difference of the electron numbers of the two states) and the amplitudes $\delta^2 = (c+m)/2$, $\epsilon^2 = (c-m)/2$ are known in closed form only at $\Delta V = 0$ ($m = 0$, $\delta = \epsilon = \sqrt{1/4 - U/32t}$ for $U \le 8t$), in the limit $U \to 0$ (above), and as $U \to U_c$ where $\delta, \epsilon \to 0$ with $m/c \to \Delta V/\sqrt{4t^2 + \Delta V^2}$; for all other $(U, \Delta V)$ they are given parametrically by the two equations above and obtained numerically from them. The critical U_c/t is in closed form for all ΔV.

Table 1 gives the main quantities.

	$\Delta V = 0$ (closed form)			$\Delta V = 2$ (two-variable reduction)		
U/t	Z	E_{var}/t	P_{ion}	Z	E_{var}/t	P_{ion}
2	0.938	−3.125	0.375	0.993	−4.774	0.781
4	0.750	−4.500	0.250	0.926	−5.422	0.558
6	0.438	−6.125	0.125	0.724	−6.549	0.321
7	0.234	−7.031	0.062	0.567	−7.280	0.220
8	0	−8.000	0	0.378	−8.105	0.130
10	0	−10.000	0	0	−10.000	0

Table 1: $N_g = 0$: renormalization factor, energy and ionic weight.

2 Total energy and error vs. exact

Figures 1 and 2 show the variational energy E_{var}/t of the $N_g = 0$ and $N_g = 2$ solutions together with the exact (ED) energy, and the error $E_{\mathrm{var}} - E_{\mathrm{ED}}$ on a logarithmic scale, for $\Delta V = 0$ and $\Delta V = 2$ respectively. For $\Delta V = 2$ the plotted $N_g = 2$ energies are the cross-checked ("confirmed") solutions; the $N_g = 0$ energies are the closed form of Sec. 1 at $\Delta V = 0$ and its two-variable reduction at $\Delta V = 2$. The two curves agree to 10^{-3} up to $U/t = 2.3$ ($\Delta V = 0$) and 4.3 ($\Delta V = 2$); $N_g = 0$ is higher by at most 0.13 (at $U/t = 7.7$ and 9.4).

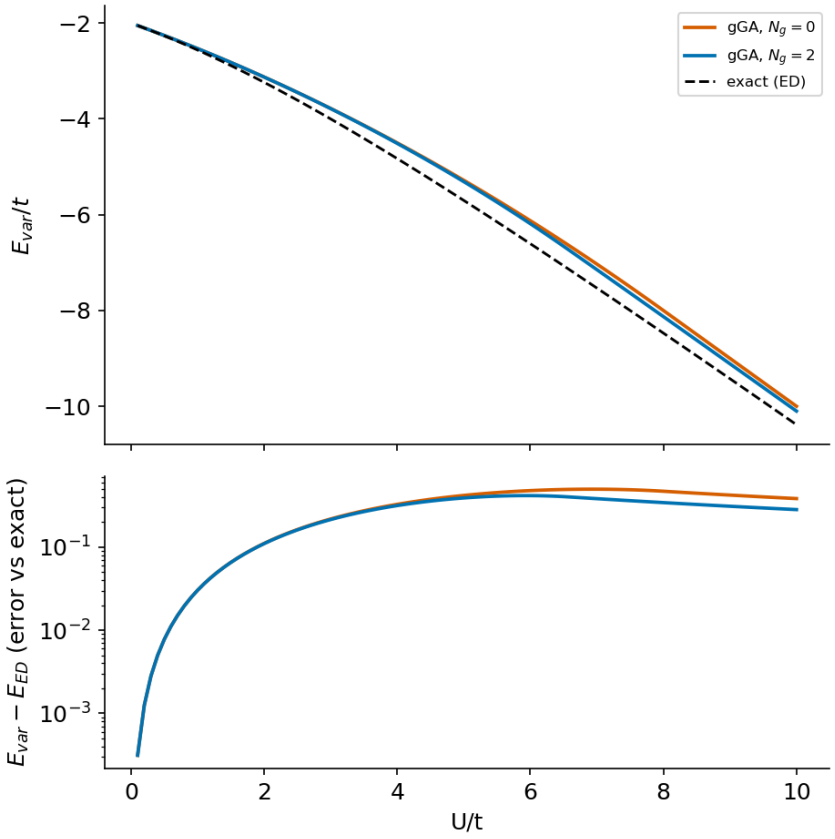


Figure 1: $\Delta V = 0$: E_{var}/t (top) and error $E_{\mathrm{var}} - E_{\mathrm{ED}}$ (bottom) vs. U/t for $N_g = 0$ and $N_g = 2$.

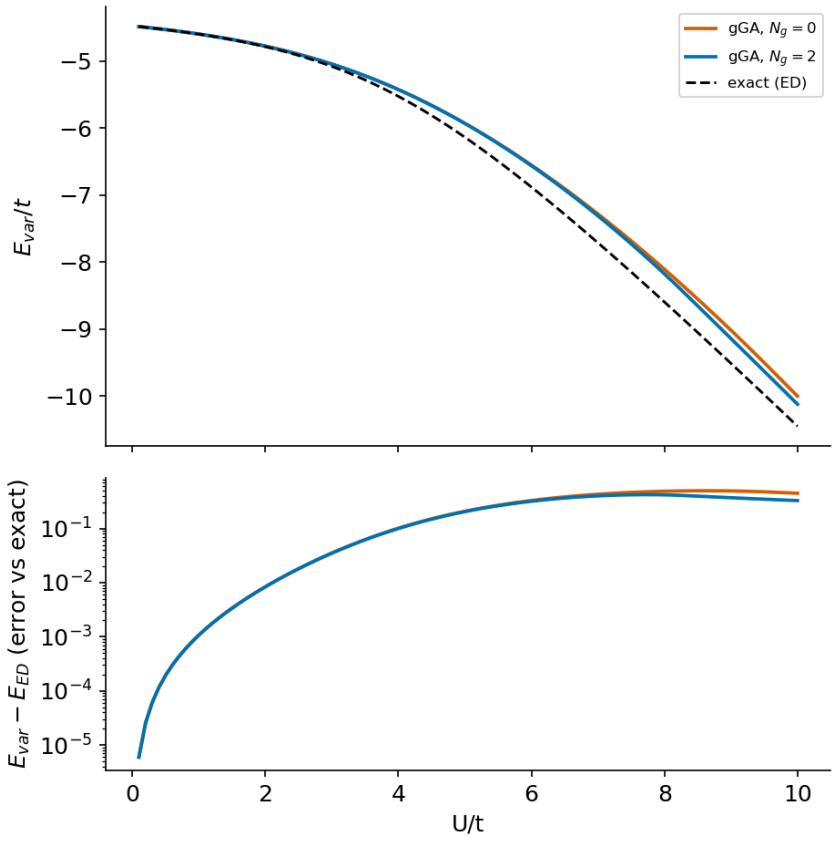
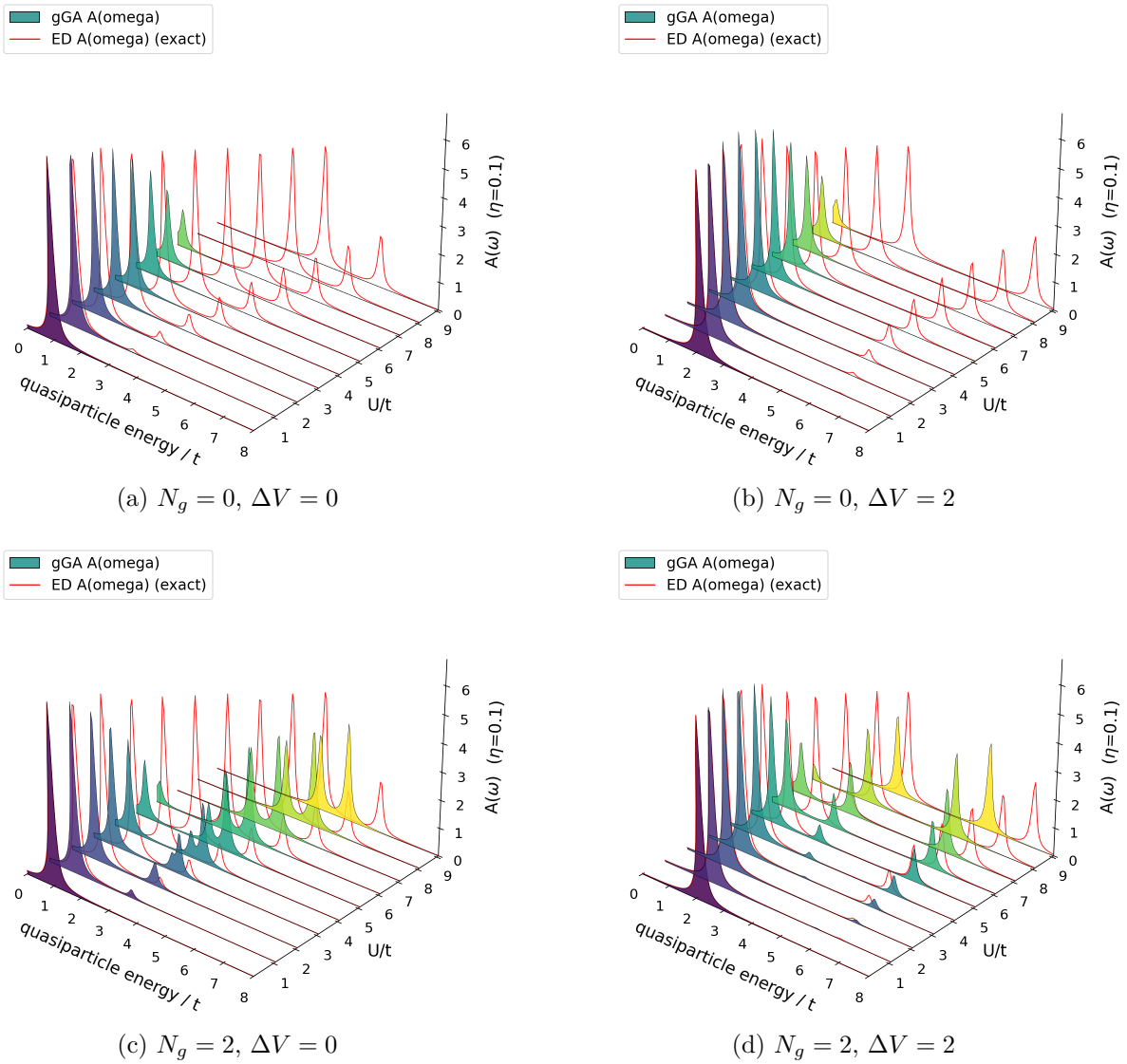


Figure 2: $\Delta V = 2$: E_{var}/t (top) and error $E_{\mathrm{var}} - E_{\mathrm{ED}}$ (bottom) vs. U/t for $N_g = 0$ and $N_g = 2$.

3 Spectra and spectral-function movies

3.1 3D view

Figure 3 shows, for each $(N_g, \Delta V)$ combination, the gGA spectral function $A(\omega)$ (filled ridgelines, built from the quasiparticle poles and weights of the converged solution, broadening $\eta = 0.1$) as a function of U/t, together with the exact (ED) spectral function (red lines). Rows are $N_g = 0$ and $N_g = 2$, columns are $\Delta V = 0$ and $\Delta V = 2$. The $N_g = 2$, $\Delta V = 2$ panel uses the cross-checked solutions.



(a) $N_g = 0$, $\Delta V = 0$

(b) $N_g = 0$, $\Delta V = 2$

(c) $N_g = 2$, $\Delta V = 0$

(d) $N_g = 2$, $\Delta V = 2$

Figure 3: gGA spectral function $A(\omega)$ (filled) vs. ED (red) as a function of U/t for $N_g = 0$ (top) and $N_g = 2$ (bottom), at $\Delta V = 0$ (left) and $\Delta V = 2$ (right). Only $\omega \geq 0$ is shown: $A(\omega) = A(-\omega)$ in both gGA and ED; the $N_g = 0$ panels are the analytic solution. Scripts: `plot_eqp_3d_ng0.py`, `plot_eqp_3d_pos.py`.

Analysis. The number of quasiparticle branches is $2(1 + N_g)$: two for $N_g = 0$ and six for $N_g = 2$. In all four panels the gGA peaks coincide with the ED ones at small U/t and move away from them as U/t grows, consistent with the error in the total energy growing with U/t (Sec. 2). For $N_g = 0$ there is a single positive peak, at tZ with weight $2Z$ at $\Delta V = 0$, which goes to $\omega = 0$ and vanishes at $U_c/t = 8.00$ (9.66 at $\Delta V = 2$): the spectral function disappears. For $N_g = 2$ the extra branches give the outer peaks, and the inner peak vanishes at the lower $U_c/t = 6.56$ (8.33 at $\Delta V = 2$). The effect of ΔV is visible for both N_g: at $\Delta V = 2$ the branches are well separated at small U/t and bunch together as U/t increases, with a rearrangement for $N_g = 2$ between $U/t = 0.6$ and 0.7 (Sec. 5).

3.2 Spectral function movies

- $N_g = 0$, $\Delta V = 0$: spectral-function movie
- $N_g = 0$, $\Delta V = 2$: spectral-function movie
- $N_g = 2$, $\Delta V = 0$: spectral-function movie
- $N_g = 2$, $\Delta V = 2$: spectral-function movie

The $N_g = 0$ movies are drawn from the analytic solution of Sec. 1. (Links open the movie with the PDF viewer's associated application, if the viewer supports `run:` links.)

Quasiparticle-weight movies. The same scans are shown level by level: for each U/t the eigenvalues ϵ_n^{qp} of H_{qp} with the weight carried by each level (summed over both sites and spins, so the total weight is that of $A(\omega)$), and the ED poles as red open circles (linear and logarithmic scales; script `make_qp_weights_video.py`). A level with weight $\geq 10^{-3}$ is a peak (levels without weight are light gray). A peak keeps its color along U/t (consecutive U/t points are matched by closeness in energy and weight), and the colors **blue**, **orange**, **green** follow the order of increasing energy at the U/t where all the peaks coexist.

- $N_g = 0$, $\Delta V = 0$: quasiparticle-weight movie
- $N_g = 0$, $\Delta V = 2$: quasiparticle-weight movie
- $N_g = 2$, $\Delta V = 0$: quasiparticle-weight movie
- $N_g = 2$, $\Delta V = 2$: quasiparticle-weight movie

Weight transfer. Positive-energy side. Each peak is written as (weight, energy) in the color of its peak in the movies (weights summed over sites and spins, total 2.0; ED poles in red). At $\Delta V = 0$ the peaks are the blue one, which carries almost all the weight at small U/t, and the orange and green ones, which are born at $U/t \simeq 0.5$ near $\epsilon = 3$. As U/t grows the blue peak loses weight to the orange and green ones while its energy goes to zero, and both vanish at a critical U_c/t (ED has no such event; the values come from a linear extrapolation of the last grid points); the orange and green peaks continue, with the same colors, above U_c.

- $\Delta V = 0$, $U/t \lesssim 1$ ($U/t = 1$): (1.92, 0.97) carries $\geq 96\%$; (0.02, 2.99) and (0.06, 3.02) hold the rest. ED: (1.97, 1.06), (0.03, 3.06).
- $\Delta V = 0$, $1 \lesssim U/t < U_c/t \simeq 6.56$: the blue peak loses its weight and the other two gain it. $U/t = 4$: (1.08, 0.57), (0.27, 2.83), (0.66, 3.26); $U/t = 6$: (0.27, 0.14), (0.66, 2.56), (1.05, 3.41); $U/t = 6.5$: (0.03, 0.02), (0.80, 2.46), (1.13, 3.43). ED at $U/t = 4$: (1.71, 1.83), (0.29, 3.83).
- $\Delta V = 0$, $U/t > U_c/t$: only the orange and green peaks. $U/t = 7$: (0.83, 2.69), (1.13, 3.67); $U/t = 10$: (0.89, 4.29), (1.10, 5.29). ED at $U/t = 10$: (1.37, 4.39), (0.63, 6.39).

At $\Delta V = 2$ the blue peak is again the main one; the orange peak is the middle one (born at $U/t \simeq 1.5$–3) and the green one the upper one (born at $U/t \simeq 1$).

- $\Delta V = 2$, $U/t \lesssim 3$ ($U/t = 3$): (1.89, 1.25), (0.015, 3.59), (0.10, 5.99); ED: (1.89, 1.34), (0.11, 5.81).
- $\Delta V = 2$, $3 \lesssim U/t < U_c/t \simeq 8.3$: the blue peak loses weight and the other two gain it. $U/t = 5$: (1.46, 0.81), (0.18, 2.57), (0.36, 5.90); $U/t = 6$: (1.10, 0.59), (0.37, 2.33), (0.53, 5.83); $U/t = 8$: (0.18, 0.09), (0.84, 2.08), (0.95, 5.28). The blue peak has weight 0.02 at $U/t = 8.3$ and 10^{-3} at 8.5 (it stays at $\epsilon \lesssim 10^{-3}$).
- $\Delta V = 2$, $U/t > U_c/t$: only the orange and green peaks. $U/t = 9$: (0.94, 2.38), (1.02, 5.66); $U/t = 10$: (0.96, 2.88), (1.02, 6.36). ED at $U/t = 10$: (1.22, 3.21), (0.78, 7.68).

Above U_c the two weights approach each other but are not equal up to $U/t = 10$; gGA is closer to equal sharing than ED, i.e. it puts too much weight on the upper level.

3.3 Quasiparticle orbitals and density: the gap closing at U_c

Two more movies per case show the eigenorbitals of H_{qp} and the quasiparticle density as U/t grows ($N_g = 0$ and 2; $\Delta V = 0, 2$; script `make_qp_orbital_videos.py`). They are drawn on the skeleton of the dimer: the two large circles joined left to right are the two states (the direction of R_s, the only qp orbital that hops to the other site), and the small outer circles are their ghosts (the on-site eigenvectors orthogonal to it). Occupation as in the code (`gga_dimer.qp_state`): the lowest $N/2$ eigenvectors returned by `eigh` are filled, without fractional occupation. In the orbital movies each panel follows one orbital as U/t grows (it is matched to the previous U/t by maximum overlap, so it does not jump when levels cross or split; the orbitals of degenerate levels are rotated onto the previous ones); the title gives its occupation, energy and $A(\omega)$ weight. Circle area is $|\text{amplitude}|^2$ and the color is the sign (the signs of the orbitals and of the ghost circles are aligned with the previous U/t, so a color change is a change of the orbital, not of the phase). In the density movies the colors give the qp density per orbital (the code's ρ), next to the gaps, bond order, Z and double occupancy (or state density at $\Delta V = 2$) as functions of U/t; the dashed line is U_c/t.

- $N_g = 0$, $\Delta V = 0$: orbitals, density
- $N_g = 0$, $\Delta V = 2$: orbitals, density
- $N_g = 2$, $\Delta V = 0$: orbitals, density
- $N_g = 2$, $\Delta V = 2$: orbitals, density

What can be seen at the critical U_c/t of the weight transfer (6.56 for $\Delta V = 0$, $\simeq 8.3$ for $\Delta V = 2$):

- *Spectral gap* (first panel of the density movies, lowest positive minus highest negative energy among the levels with weight $\geq 10^{-3}$): it decreases smoothly with U/t (2.0 at $U/t = 0.1$, 1.14 at 4, 0.27 at 6 for $\Delta V = 0$; 4.39, 2.0 (at $U/t = 4$) and 1.19 (at 6) for $\Delta V = 2$, $N_g = 2$) and closes at U_c; the next grid point has a large gap ($0.03 \to 4.94$ between $U/t = 6.5$ and 6.6 at $\Delta V = 0$; $0.001 \to 4.36$ between 8.5 and 8.6 at $\Delta V = 2$, $N_g = 2$), and it grows afterwards (at $U/t = 10$: 8.59 and 5.76). The qp gap (HOMO–LUMO) is zero from U_c on; before U_c it follows the spectral gap, except at small U/t, where weightless ghost levels lie inside the gap.
- *Orbitals*: below U_c the HOMO and LUMO have amplitude on the two states and carry $A(\omega)$ weight; from U_c on, the orbitals at $\epsilon = 0$ have no amplitude on the states and zero weight, and sit on ghost circles only (two of them). Which of these degenerate orbitals are occupied is the choice of `eigh`, so the occupied ghost orbitals change from frame to frame above U_c.
- *Bond order and double occupancy* (center–center density matrix element, per spin): both change slope at U_c (at $\Delta V = 0$: bond order 0.17, 0.09, 0.08 and double occupancy 0.040, 0.016, 0.013 at $U/t = 6$, 6.5, 6.6). At $\Delta V = 2$ the bond order has a maximum (0.40) at $U/t \simeq 4.3$ and falls to 0.08 at U_c.
- *State density at* $\Delta V = 2$: the densities of the two states are polarized at small U/t (0.945 and 0.055 at $U/t = 0.1$, 0.63 and 0.37 at 6) and become equal at U_c (0.50 and 0.50, $N_g = 2$), staying equal above.
- Z stays ≥ 0.975 for all U/t (no collapse of the renormalization), and at $\Delta V = 0$ the state densities are 0.50 at all U/t; the colors of the ghost circles are not a signature of the transition.

This is a gap closing and re-opening of the gGA solution. It is not present in the exact dimer, whose spectral gap varies smoothly with U/t (Sec. 3).

3.4 $N_g = 0$ orbitals versus the $N_g = 2$ HOMO and LUMO

Relation between the two. At each U/t the part of the $N_g = 2$ HOMO and LUMO that sits on the two states has the same shape as the $N_g = 0$ occupied and virtual orbital: the same relative amplitudes and signs (the global signs of the $N_g = 0$ orbitals are chosen to match). The rest of the $N_g = 2$ orbitals sits on the ghosts; it takes weight away from the state part, so the weight of the $N_g = 2$ HOMO and LUMO vanishes at a lower U_c/t than the $N_g = 0$ one (6.56 and 8.33 against 8.00 and 9.66 at $\Delta V = 0$ and 2). The transition does *not* take place at the same U/t, but the orbitals are the same.

Table 2 gives, for the $N_g = 2$ HOMO, the weight n_s on the two states, the overlap between its state part and the $N_g = 0$ occupied orbital, the level energies and the $A(\omega)$ weights of both (the LUMO has the same values with the opposite energy; script `ng0_vs_ng2_orbitals.py`). The overlap is 1.000 over the whole range (0.996 at $U/t = 8$, $\Delta V = 2$, where n_s is already 0.09), n_s falls to zero at U_c, and the weight is $A \simeq 2n_s$ for $N_g = 2$ against $2Z$ for $N_g = 0$. At $\Delta V = 2$ and $U/t \lesssim 1.5$ the $N_g = 2$ HOMO and LUMO are weightless ghost levels, and the orbitals that carry the state weight are the deeper ones.

	$N_g = 2$ HOMO: state part		$\lvert\epsilon\rvert/t$		$A(\omega)$ weight	
U/t	n_s	overlap with $N_g = 0$	$N_g = 2$	$N_g = 0$	$N_g = 2$	$N_g = 0$
$\Delta V = 0$						
2	0.86	1.000	0.88	0.94	1.72	1.88
4	0.54	1.000	0.57	0.75	1.08	1.50
5	0.35	1.000	0.36	0.61	0.69	1.22
6	0.13	1.000	0.14	0.44	0.27	0.88
$\Delta V = 2$						
2	0.98	1.000	1.52	1.54	1.96	1.99
4	0.86	1.000	1.02	1.08	1.73	1.85
6	0.55	1.000	0.59	0.75	1.09	1.45
8	0.09	0.996	0.09	0.38	0.18	0.76

Table 2: $N_g = 2$ HOMO against the $N_g = 0$ occupied orbital: weight n_s on the two states, overlap of the state part with the $N_g = 0$ orbital, level energy and $A(\omega)$ weight.

3.5 Ionic and covalent weights

As U/t grows (the analogue of stretching H_2) the weight of the ionic configurations (one site doubly occupied, H^-H^+) is expected to go to the covalent ones (one electron on each site). The ionic weight is $P_{\text{ion}} = \langle n_{0\uparrow} n_{0\downarrow} \rangle + \langle n_{1\uparrow} n_{1\downarrow} \rangle$ and the covalent weight is $1 - P_{\text{ion}}$. In gGA the double occupancies and the site occupations come from the impurity ground state of each fragment (`gga_dimer.impurity_solve`) at the converged (R, λ), as in one iteration of the loop (at $\Delta V = 0$ they reproduce the `docc` of the solver output); ED uses the exact ground state ($\Delta V_{\text{ED}} = 2\,\Delta V_{\text{gGA}}$). Figure 4 gives P_{ion} over the whole U/t grid for both ΔV.

	$\Delta V = 0$: P_{ion}			$\Delta V = 2$: P_{ion}			$\Delta V = 2$: n favored site	
U/t	ED	$N_g = 0$	$N_g = 2$	ED	$N_g = 0$	$N_g = 2$	ED	$N_g = 2$
0.1	0.488	0.494	0.494	0.896	0.896	0.896	1.890	1.890
2	0.276	0.375	0.374	0.767	0.781	0.781	1.755	1.764
4	0.146	0.250	0.238	0.469	0.558	0.557	1.446	1.517
6	0.084	0.125	0.079	0.205	0.321	0.310	1.178	1.261
7	0.066	0.062	0.022	0.138	0.220	0.193	1.113	1.154
8.5	0.048	0	0.015	0.083	0.089	0.028	1.061	1.021
10	0.036	0	0.010	0.055	0	0.017	1.036	1.011

Table 3: Ionic weight (ED and gGA, $N_g = 0$ and 2) and, at $\Delta V = 2$, electrons on the favored site ($n = 1$ is purely covalent).

At $\Delta V = 0$ there is no charge transfer ($n = 1$ on each site) and the ionic weight falls from 0.5, i.e. the weight moves from ionic to covalent, in ED smoothly (0.036 at $U/t = 10$). $N_g = 0$ falls linearly and vanishes at $U_c/t = 8$; it agrees with $N_g = 2$ up to $U/t \simeq 2$ and is more ionic for $2 \lesssim U/t \lesssim 6$ (0.125 against 0.079 at $U/t = 6$). gGA ($N_g = 2$) does the same but is too ionic at intermediate U/t and then drops across $U_c/t \simeq 6.56$ (0.079 $\to$ 0.022 between $U/t = 6$ and 7) to an almost purely covalent state (0.010 at $U/t = 10$). At $\Delta V = 2$ the state is polarized (H^-H^+-like, 1.89 electrons on the favored site) and becomes covalent as U/t grows; gGA is more ionic than ED up to $U/t \simeq 7$ and less ionic after $U_c/t \simeq 8.3$ ($n = 1.021$ against 1.061 at $U/t = 8.5$). The density of the two states in the density movies matches the physical occupation per spin at the two points checked (0.945 at $U/t = 0.1$ and 0.630 at 6 for $\Delta V = 2$).

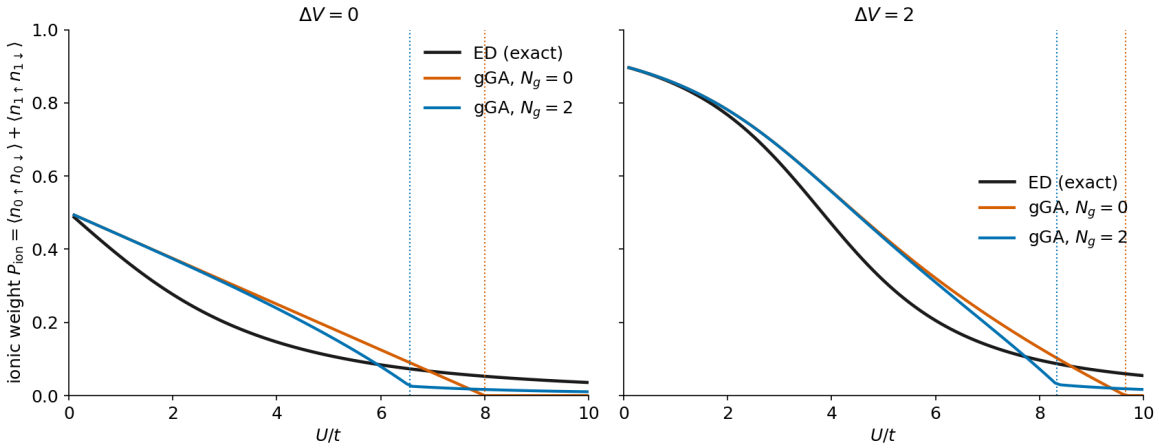


Figure 4: Ionic weight P_{ion} of the ground state vs. U/t: ED (black), gGA with $N_g = 0$ (orange) and $N_g = 2$ (blue), for $\Delta V = 0$ (left) and $\Delta V = 2$ (right); the dotted lines are U_c/t of $N_g = 0$ (8.00 and 9.66) and of $N_g = 2$ (6.56 and 8.34). For $N_g = 0$ the curve is the closed form of Sec. 1 at $\Delta V = 0$ and its two-variable reduction at $\Delta V = 2$. For $N_g = 2$ and $\Delta V = 0$ the gGA values are $2\,$`docc` from the solver output; at $\Delta V = 2$ they are the impurity double occupancies $\sum_I \phi_I^T N_{dd}\, \phi_I$ of the embedding states. Script: `make_ng0_figures.py`.

P_{ion} from the Gutzwiller wavefunction. In the gGA a local expectation value in the Gutzwiller wavefunction is the one in the embedding state, so $P_{\text{ion}} = \sum_I \phi_I^T N_{dd}\, \phi_I$ can be read from the Φ-matrix of the direct minimization, without the impurity solve. At a solution it equals the impurity

value of Fig. 4 (script `pion_check_refined.py`).

4 Critical U_c/t versus ΔV

Figure 5 follows the transition of the gGA solution as the site splitting grows from $\Delta V = 0$ to 5 in steps of 0.2, for $N_g = 0$ and $N_g = 2$. At each ΔV the direct-minimization solution is followed in U/t (steps of 0.1). For $N_g = 2$ it starts from the $\Delta V = 0$ solution and is continued in ΔV, and U_c/t is the value where the spectral gap (Sec. 3) goes to zero, from a linear extrapolation of the last grid points before it closes (script `Uc_from_scan.py`, data in `scan_Uc_vs_dV/`). For $N_g = 0$ the spectral function disappears when $Z \to 0$, and $U_c/t = 4 + 2\sqrt{4 + (\Delta V/t)^2}$ (Sec. 1); the extrapolation of Z to zero of the numerical solutions (script `ng0_analysis.py`, data in `scan_ng0/`) agrees with it to 1.3×10^{-3} at all ΔV. The values at $\Delta V = 0$ and 2 reproduce those of Secs. 1 and 3 (8.00 and 9.66 for $N_g = 0$; 6.56 and 8.33 for $N_g = 2$).

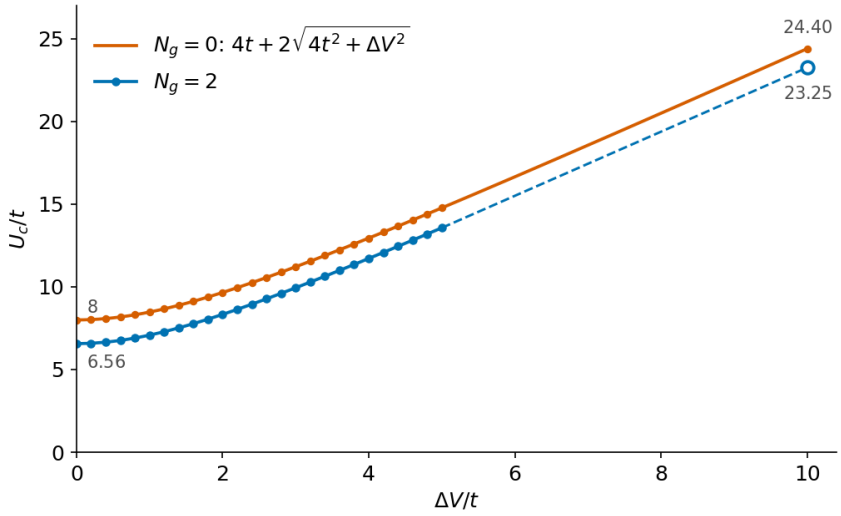


Figure 5: Critical U_c/t versus ΔV for $N_g = 0$ (orange; curve: closed form, dots: numerical solutions) and $N_g = 2$ (blue); the open circle is the single point at $\Delta V = 10$ of $N_g = 2$, computed separately, and the dashed line is a straight segment joining it to $\Delta V = 5$ (no points in between). Script: `make_ng0_figures.py`.

$\Delta V/t$	0	1	2	3	5	10
U_c/t, $N_g = 0$	8.00	8.47	9.66	11.21	14.77	24.40
U_c/t, $N_g = 2$	6.56	7.08	8.33	9.94	13.57	23.25

Table 4: Critical U_c/t for $N_g = 0$ (closed form) and $N_g = 2$.

For both N_g, U_c/t increases monotonically with ΔV: it is flat at small ΔV ($N_g = 2$: 6.56, 6.59, 6.65 at $\Delta V = 0$, 0.2, 0.4) and grows almost linearly for $\Delta V \gtrsim 3$ (slope $\simeq 1.8$ for both; mean slopes 1.93 for $N_g = 0$ and 1.94 for $N_g = 2$ between $\Delta V = 5$ and 10; the $N_g = 0$ closed form tends to slope 2). $N_g = 0$ lies above $N_g = 2$ at all ΔV, by 1.44 at $\Delta V = 0$ and 1.14 at $\Delta V = 10$: the ghost orbitals lower U_c.

5 Discussion: $\Delta V = 0$ vs. $\Delta V = 2$

Weak coupling: ΔV dominates. At small U/t the two scans differ substantially. At $U/t = 0.1$, $E_{\text{var}}/t = -2.0503$ for $\Delta V = 0$ but -4.4823 for $\Delta V = 2$. This is the expected $U \to 0$ limit: with $\Delta V = 0$ the dimer's noninteracting energy is $-2t = -2$ plus a small $O(U)$ correction, while at $\Delta V = 2$ the site-splitting opens a one-body level gap and the $U \to 0$ energy approaches the closed form $E = -2\sqrt{t^2 + \Delta V^2} = -2\sqrt{5} \approx -4.472$. The gap between the two curves is largest here, $\Delta E \approx 2.43$, because ΔV acts as an unscreened one-body potential that the (still weak) correlations barely renormalize.

Strong coupling: the curves converge. As U/t grows the two energies draw close together: at $U/t = 10$, $E_{\text{var}}/t = -10.101$ ($\Delta V = 0$) vs. -10.120 ($\Delta V = 2$), a difference of only ≈ 0.02 – two orders of magnitude smaller than at weak coupling. In the Mott-like regime the total energy is governed almost entirely by U (both sites approach single occupancy), and the fixed $\Delta V = 2$ site-splitting becomes a small perturbation on top of the dominant Hubbard physics. The quasiparticle spectra (3D plots, Fig. 3) show the same trend: the $\Delta V = 2$ branches, well separated at small U, bunch together with the $\Delta V = 0$-like structure as U/t increases.

Level crossing at $\Delta V = 2$. The $\Delta V = 2$ quasiparticle energies show a sharp rearrangement between $U/t = 0.6$ and $U/t = 0.7$ (e.g. $N_g = 2$: $\epsilon^{qp} \approx -2.00 \to -3.12$), visible as a kink in the 3D plots; this is the level crossing where the ΔV-induced one-body gap stops being the dominant scale and the Hubbard repulsion takes over.

Ghost multiplicity ($N_g = 0$ vs. $N_g = 2$). The ghost orbitals lower the energy only at larger U/t: $E(N_g{=}0) - E(N_g{=}2) < 10^{-3}$ up to $U/t = 2.3$ at $\Delta V = 0$ and 4.3 at $\Delta V = 2$, and at most 0.13 in the scanned range (Sec. 2), still well below the error to the exact energy at large U/t (≈ 0.3–0.5). The ghosts change the transition more than the energy: U_c/t goes from 8 ($N_g = 0$) to 6.56 ($N_g = 2$) at $\Delta V = 0$ and the transition is no longer a continuous vanishing of Z (the $N_g = 0$ spectral function disappears with Z, the $N_g = 2$ one keeps $Z \geq 0.975$ and loses its inner peak, Sec. 3).

6 The Gutzwiller wave function itself

gGA never forms the state it describes. Here it is built explicitly from the converged solutions (script `gutzwiller_psi.py`; scans `psi_scan.py`, data in `psi_scan/`), to read off the ionic and covalent content, the energy and the overlap with the exact ground state (FCI, from `exact_dimer_deltaV.py`).

Construction. For each fragment, $|\Psi\rangle = P_0 P_1 |\Psi_{\text{qp}}\rangle$, with $|\Psi_{\text{qp}}\rangle$ the Slater determinant of the occupied levels of H_{qp} and $P_i = \sum_{\Gamma n} \lambda_{\Gamma n} |\Gamma\rangle\langle n|$ mapping the qp Fock space of the fragment ($4^{N_{\text{eff}}}$ states, $N_{\text{eff}} = 1 + N_g$) onto its four physical configurations Γ. The matrix is $\lambda = \tilde{\Phi}\, P^{0\,-1/2}$, with P^0 the local density matrix of Ψ_{qp} and $\tilde{\Phi}$ the embedding state Φ with the bath particle-hole transformed back, $\tilde{\Phi} = \Phi\, U_{\text{PH}}$, $U_{\text{PH}}|n\rangle = (-1)^{N(N-1)/2} f_{j_1} \cdots f_{j_N} |\text{full}\rangle$. The phase fixes the relative sign of the covalent and ionic amplitudes (the other choice gives $E_{\text{kin}} > 0$). Checks: $R = S^{-1}\text{Tr}[\tilde{\Phi}^T c^\dagger \tilde{\Phi} f]$ reproduces the R of the solution to 10^{-7} (fully for the non-impurity points), and the weight of $\tilde{\Phi}$ outside the support of P^0 is $\leq 7 \times 10^{-8}$. Φ is the saved ϕ_i for non-impurity solutions (`z.npy`) and the ground state of H_{imp} for impurity solutions. $N_g = 2$, $\Delta V = 2$ uses the non-impurity scan (`scan_ng2_not_imp_dV2`): the impurity-solver points (`scan_ng2_dV2_confirmed`)

have the same energies, but at $U/t = 0.6, 0.7, 9.9, 10$ the Fermi level is degenerate or the ghosts are inert, $\Psi_{\rm qp}$ does not match Φ, and Ψ fails there.

Normalization and energy. The gGA constraints give $\langle\Psi|\Psi\rangle = 1$ only in infinite dimensions; here it grows from 1 to $\simeq 2.8$ with U/t (Tables 5, 6; 0.29 for $N_g = 2$, $\Delta V = 2$ at small U/t, where the ghosts are inert and the value depends on the gauge). All quantities below use the normalized Ψ. The energy $E_\Psi = \langle\Psi|H|\Psi\rangle$ is variational and far better than the gGA one: for $N_g = 2$ the error is $\leq 8.5 \times 10^{-4}$ over the whole scan (against up to 0.4 for gGA) and the overlap with FCI is ≥ 0.9999. For $N_g = 0$ it is 0.03 at $U/t = 4$ and 0.3 at 7 (overlap 0.98 at U_c); $N_g = 0$ at $\Delta V = 0$ is the ordinary Gutzwiller wave function, and gGA's E is its Gutzwiller-approximation value. Above U_c of $N_g = 0$ ($Z = 0$, $R = 0$) the solution does not determine $\Psi_{\rm qp}$: the values marked † (dashed in Fig. 6) depend on the arbitrary $\Psi_{\rm qp}$ stored by the solver (e.g. $E_\Psi = -U$ and $P_{\rm ion} = 0$ exactly) and are not a result.

	$N_g = 0$				$N_g = 2$			
U/t	$\langle\Psi\|\Psi\rangle$	$E_{\rm gGA} - E$	$E_\Psi - E$	$1 - \|\langle\Psi\|{\rm FCI}\rangle\|$	$\langle\Psi\|\Psi\rangle$	$E_{\rm gGA} - E$	$E_\Psi - E$	$1 - \|\langle\Psi\|{\rm FCI}\rangle\|$
0.1	1.000	0.000	0	0	1.000	0.000	0	0
2	1.062	0.111	7.7×10^{-4}	8.7×10^{-5}	1.153	0.110	5.4×10^{-5}	6.0×10^{-6}
4	1.250	0.328	2.8×10^{-2}	2.5×10^{-3}	1.758	0.317	1.5×10^{-5}	1.3×10^{-6}
6	1.562	0.481	1.7×10^{-1}	1.2×10^{-2}	2.647	0.418	2.3×10^{-4}	1.6×10^{-5}
7	1.766	0.500	3.0×10^{-1}	1.9×10^{-2}	2.731	0.384	2.4×10^{-4}	1.5×10^{-5}
8.5	$2.000^\dagger$	$0.447^\dagger$	$4.5\times10^{-1\dagger}$	$2.4\times10^{-2\dagger}$	2.466	0.327	1.2×10^{-4}	6.3×10^{-6}
10	$2.000^\dagger$	$0.385^\dagger$	$3.9\times10^{-1\dagger}$	$1.8\times10^{-2\dagger}$	2.325	0.284	6.1×10^{-5}	2.8×10^{-6}

Table 5: $\Delta V = 0$: norm, error of the gGA energy $E_{\rm gGA} - E_{\rm FCI}$ and of $E_\Psi - E_{\rm FCI}$, and overlap deficit of Ψ with the FCI ground state. †: above U_c of $N_g = 0$, $\Psi_{\rm qp}$ is not determined.

	$N_g = 0$				$N_g = 2$			
U/t	$\langle\Psi\|\Psi\rangle$	$E_{\rm gGA} - E$	$E_\Psi - E$	$1 - \|\langle\Psi\|{\rm FCI}\rangle\|$	$\langle\Psi\|\Psi\rangle$	$E_{\rm gGA} - E$	$E_\Psi - E$	$1 - \|\langle\Psi\|{\rm FCI}\rangle\|$
0.1	1.000	0.000	0	0	0.294	0.000	0	0
2	1.003	0.008	2.1×10^{-4}	2.0×10^{-5}	0.294	0.008	2.1×10^{-4}	2.0×10^{-5}
4	1.043	0.101	6.8×10^{-3}	5.9×10^{-4}	1.106	0.100	3.1×10^{-4}	5.0×10^{-5}
6	1.216	0.329	3.4×10^{-2}	1.6×10^{-3}	1.651	0.319	1.2×10^{-4}	1.6×10^{-5}
7	1.372	0.430	7.9×10^{-2}	4.8×10^{-3}	2.099	0.402	1.7×10^{-4}	2.0×10^{-5}
8.5	1.689	0.499	2.4×10^{-1}	1.6×10^{-2}	2.700	0.397	6.3×10^{-4}	5.7×10^{-5}
10	$2.060^\dagger$	$0.449^\dagger$	$4.5\times10^{-1\dagger}$	$5.0\times10^{-1\dagger}$	2.426	0.328	2.2×10^{-4}	1.6×10^{-5}

Table 6: As Table 5 for $\Delta V = 2$ ($N_g = 2$ from the non-impurity scan).

Ionic → covalent. Table 7 compares $P_{\rm ion}$ of ED, of Ψ ($\Psi_{N_g=0}$, $\Psi_{N_g=2}$) and of the fragment ground states used so far (frag, Sec. 3). The fragments are too ionic up to $U/t \simeq 5$ ($\text{frag}_{N_g=2}$: 0.238 against 0.146 at $U/t = 4$) and too covalent from 6 on (0.079 and 0.022 against 0.084 and 0.066 at 6 and 7), while $\Psi_{N_g=2}$ follows ED within 0.006 at $\Delta V = 0$ and 0.011 at $\Delta V = 2$ (0.1453 against 0.1464 at $U/t = 4$) and keeps the slow decrease above U_c (0.037 against 0.036 at 10). $\Psi_{N_g=0}$ is too covalent: 0.100 at $U/t = 4$ and 0.020 at 6, going to 0 at U_c.

	$\Delta V = 0$: $P_{\rm ion}$					$\Delta V = 2$: $P_{\rm ion}$				
U/t	ED	$\Psi_{N_g=0}$	$\Psi_{N_g=2}$	$\mathrm{frag}_{N_g=0}$	$\mathrm{frag}_{N_g=2}$	ED	$\Psi_{N_g=0}$	$\Psi_{N_g=2}$	$\mathrm{frag}_{N_g=0}$	$\mathrm{frag}_{N_g=2}$
0.1	0.4875	0.4875	0.4875	0.4938	0.4937	0.8961	0.8961	0.8961	0.8962	0.8962
2	0.2764	0.2647	0.2733	0.3750	0.3737	0.7669	0.7702	0.7702	0.7808	0.7808
4	0.1464	0.1000	0.1453	0.2500	0.2385	0.4693	0.4888	0.4783	0.5583	0.5572
6	0.0840	0.0200	0.0871	0.1250	0.0794	0.2052	0.1822	0.2093	0.3211	0.3096
7	0.0659	0.0044	0.0686	0.0625	0.0223	0.1379	0.0847	0.1419	0.2201	0.1928
8.5	0.0476	$0.0000^{\dagger}$	0.0491	$0.0000^{\dagger}$	0.0147	0.0827	0.0128	0.0884	0.0889	0.0282
10	0.0358	$0.0000^{\dagger}$	0.0367	$0.0000^{\dagger}$	0.0104	0.0545	$0.0000^{\dagger}$	0.0570	$0.0000^{\dagger}$	0.0167

Table 7: Ionic weight $P_{\rm ion}$: ED, $\Psi_{N_g=0}$, $\Psi_{N_g=2}$ and fragment ground states (frag). †: $\Psi_{\rm qp}$ not determined (above U_c of $N_g = 0$).

At $\Delta V = 0$, $|\Psi\rangle = \cos\theta|\mathrm{cov}\rangle + \sin\theta|\mathrm{ion}\rangle$ with the amplitudes of Table 8. $\Psi_{N_g=2}$ has the exact amplitudes to 0.01 over the scan and the same relative sign. The last columns give $(\cos\theta + \sin\theta)^2$, the weight of the bonding (inner) cation state in the exact interference relation, evaluated with each amplitude pair (the final states are the exact cation eigenstates, so this is a property of the ground state, not a gGA spectrum): 1.7048 ($\Psi_{N_g=2}$), 1.7071 (ED) and 1.6000 ($\Psi_{N_g=0}$) at $U/t = 4$ and 1.5640 ($\Psi_{N_g=2}$), 1.5547 (ED), 1.2800 ($\Psi_{N_g=0}$) at 6; the lowest peak of the gGA spectrum has 1.08 at $U/t = 4$ and 0.27 at 6. So the ionic→covalent transfer is reproduced by $\Psi_{N_g=2}$ but not by the gGA spectrum: the weight carried by R and H_{qp} does not follow the composition of the state it comes from.

	ED		$\Psi_{N_g=2}$		$\Psi_{N_g=0}$		$(\cos\theta + \sin\theta)^2$		
U/t	$\cos\theta$	$\sin\theta$	$\cos\theta$	$\sin\theta$	$\cos\theta$	$\sin\theta$	ED	$\Psi_{N_g=2}$	$\Psi_{N_g=0}$
0.1	0.7159	0.6982	0.7159	0.6982	0.7159	0.6982	1.9997	1.9997	1.9997
2	0.8507	0.5257	0.8525	0.5228	0.8575	0.5145	1.8944	1.8913	1.8824
4	0.9239	0.3827	0.9245	0.3812	0.9487	0.3162	1.7071	1.7048	1.6000
6	0.9571	0.2898	0.9554	0.2952	0.9899	0.1414	1.5547	1.5640	1.2800
7	0.9665	0.2567	0.9651	0.2619	0.9978	0.0665	1.4961	1.5056	1.1327
8.5	0.9759	0.2182	0.9751	0.2216	$1.0000^{\dagger}$	0.0000	1.4258	1.4322	$1.0000^{\dagger}$
10	0.9820	0.1891	0.9815	0.1914	$1.0000^{\dagger}$	0.0000	1.3714	1.3758	$1.0000^{\dagger}$

Table 8: $\Delta V = 0$: covalent and ionic amplitudes of the exact ground state and of $\Psi_{N_g=2}$, $\Psi_{N_g=0}$ (normalized, global sign with $\cos\theta > 0$), and $(\cos\theta + \sin\theta)^2$. †: $\Psi_{\rm qp}$ not determined (above U_c).

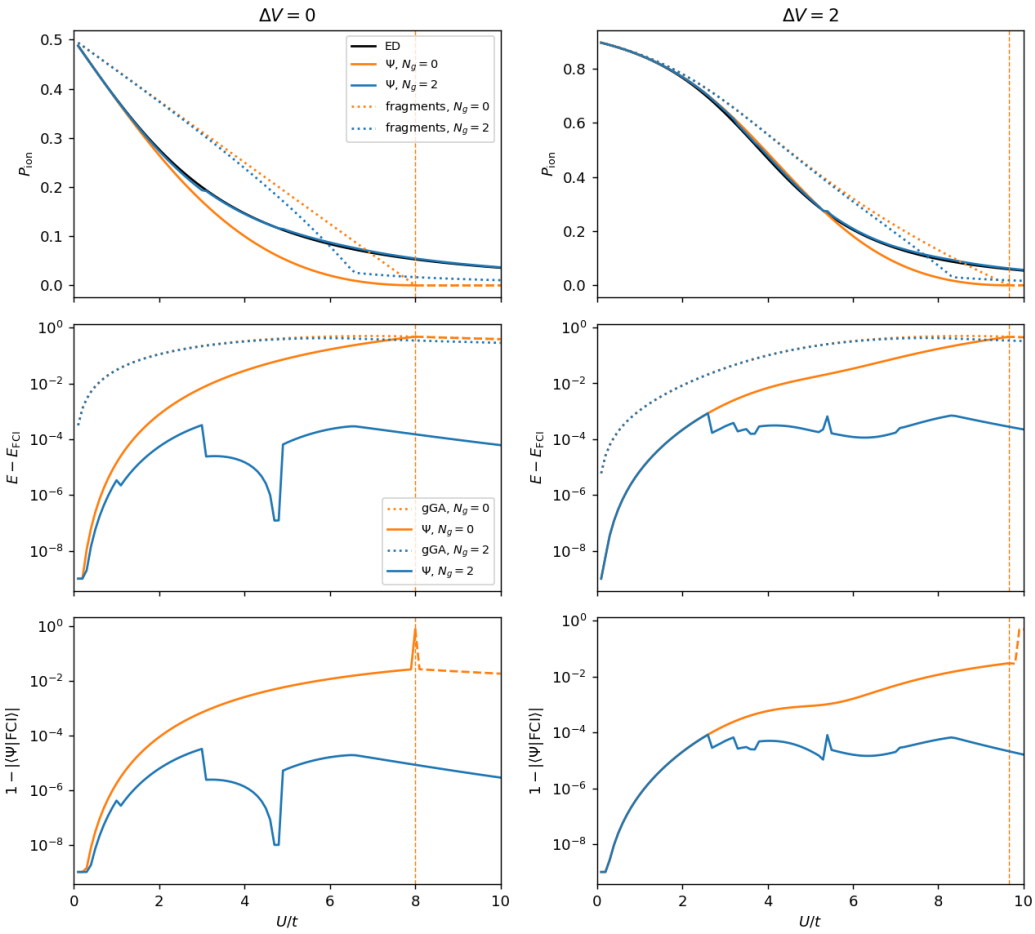


Figure 6: Whole scans. Top: P_{ion} (ED, Ψ, fragments). Middle: $E - E_{\text{FCI}}$ for gGA (dotted) and Ψ. Bottom: $1 - |\langle\Psi|\text{FCI}\rangle|$. $N_g = 0$ is dashed above its U_c/t (8.00, 9.66; vertical lines), where Ψ_{qp} is not determined.